

Tochukwu Emmanuel Okwudire

tokwudire@gmail.com | 678-328-0292 | www.linkedin.com/in/tochukwu-okwudire-6b8956290 |
<https://github.com/Toochistuff/ProjectsForResume>

EDUCATION

Kennesaw State University | Marietta, GA

August 2021 - December 2025

Bachelors of Science in Computer Science with Concentration in Cybersecurity

- Relevant Coursework: Data Structures, Computer Organization & Architecture, Operating Systems, Algorithm Analysis, Computer Networks, Cryptography, Object- Oriented Programming, Cloud Software Development, Parallel and Distributed Computing

Professional Organizations: National Society of Black Engineers (Member), Africans Students Association (Member)

Extracurriculars: Participant of Peach Hackathon and KSU Hackathon

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, HTML, CSS, Powershell

Frameworks/Libraries: Matplotlib, NumPy, Express.js, React, MySQL, JavaFX, Sockets

Tools: Github, Visual Studios, IntelliJ, Linux, PyCharm, WebStorm, VS Code, Oracle VisualBox, Ubuntu, VMware, AWS

Methodologies: Agile, Test-Driven Development, CI/CD

PROJECTS

KSU - Linux Anomaly Detection System

Undergraduate Student

Fall 2025

- Designed and implemented an end to end log analysis pipeline that gathers systemd journal logs, parses and structures raw data, and extracts numerical and categorical features for machine learning based anomaly detection.
- Built an anomaly detection module trained on curated datasets to identify suspicious behavior such as excessive sudo usage, repeated failed login attempts and abnormal access times within a Linux environment
- Developed a real time alerting system that validates model output and generates specific actionable security alerts with testing performed via simulated log events and system logger commands.

KSU - End to End Encrypted Messaging Application

Undergraduate Student

Spring 2025

- Built a secure, peer-to-peer messaging app with support for text, image, and voice messages over a local network.
- Developed a user-friendly JavaFX interface for intuitive message sending, with future media message support. It also implemented AES encryption to ensure confidentiality and used the CIA triad (Confidentiality, Integrity, Availability) as a baseline for security features.
- Implemented the voice messaging feature using Java Sockets and AES encryption, ensuring secure media transmission across clients.

KSU - Efficiency of String Algorithms

Undergraduate Student

Fall 2024

- Collaborated on testing the efficiencies when it came to 3 different string matching algorithms. (KMP, Rabin-Karp, and Boyer-Moore)
- Developed Java programs to test preprocessing time, pattern frequency, and asymptotic behavior using real-world text datasets ,performance metrics and runtime graphs to compare algorithms across large-scale data (2.5M–500M characters).
- Concluded KMP had fastest average performance due to linear preprocessing, while Boyer-Moore excelled in skipping unmatched sections via heuristics

JOB EXPERIENCE

Home Depot– Hiram, GA

Sales Associate

September 2021 – Current

- Delivered knowledgeable and customer-focused service, assisting over 50+ customers daily in finding tools, hardware, and home improvement products.
- Collaborated with team members to troubleshoot customer issues and resolve product inquiries, demonstrating strong problem-solving and communication skills.
- Took initiative to learn internal inventory systems and digital tools, building a foundation for transitioning into tech and software-focused roles.

